Big Data Analytics Lab: R Studio and R Markdown

Dom Richards

September 26, 2025

This document covers the installation and basic usage of R, RStudio, and R Markdown.

1 Introduction

R is a programming language and environment for statistical computing. RStudio is an integrated development environment (IDE) that makes working with R more efficient and user-friendly. R Markdown is a file format that allows you to create dynamic documents that combine code, results, and narrative text in a single document (similar to Jupyter notebooks).

You'll now be guided through setting up your R and RStudio environment.

2 Installing R and RStudio

2.1 Install R

R must be installed before RStudio, as RStudio is simply an interface for R.

1. Visit the Comprehensive R Archive Network (CRAN) at https://cran.r-project.org/ (figure 1)
2. Click on the appropriate link for your operating system:
   - **Windows**: Click "Download R for Windows" → "base" → "Download R 4.x.x for Windows"
   - **macOS**: Click "Download R for (Mac) OS X" → Download the appropriate .pkg file
   - **Linux**: Click "Download R for Linux" and follow the instructions for your distribution
3. Run the installer and follow the installation wizard with default settings

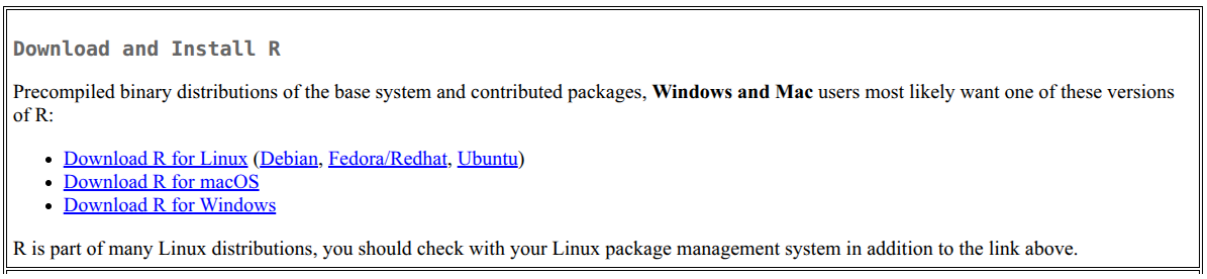


Figure 1: R installation page

2.2 Install RStudio Desktop

1. Visit https://posit.co/download/rstudio-desktop/
2. Scroll down and click on the correct version for your operating system (figure 2)
3. Run the installer and follow the installation wizard
4. Launch RStudio to verify the installation

Note: Ensure you install RStudio *after* installing R, as RStudio requires R to function.

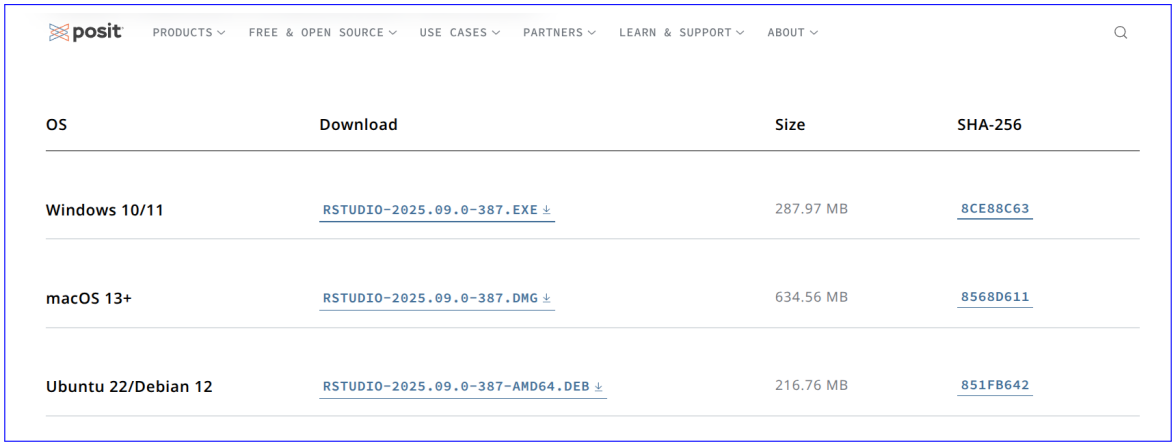


Figure 2: RStudio Desktop installation page

3 Exploring the RStudio Interface

Once RStudio is installed and running, familiarize yourself with the interface (figure 3). RStudio is divided into four main panes:

Console (bottom-left) Where R code is executed interactively

Source/Editor (top-left) Where you write and edit R scripts and R Markdown files

Environment/History (top-right) Shows variables in your workspace and command history

Files/Plots/Packages/Help (bottom-right) File browser, plot viewer, package manager, and help documentation

4 Running an R Markdown Document

Download `big_data_labs_week_2_questions.Rmd`, and use the file browser (bottom-right pane) to open it.

4.1 Understanding R Markdown Structure

The `Rmd` file contains three main components, as you can see in the top left pane (figure 3):

YAML Header Metadata at the top between `---` markers

Markdown Text Regular text with formatting using Markdown syntax

Code Chunks R code enclosed in ```` ```{r} ```` and ```` ``` ```` markers

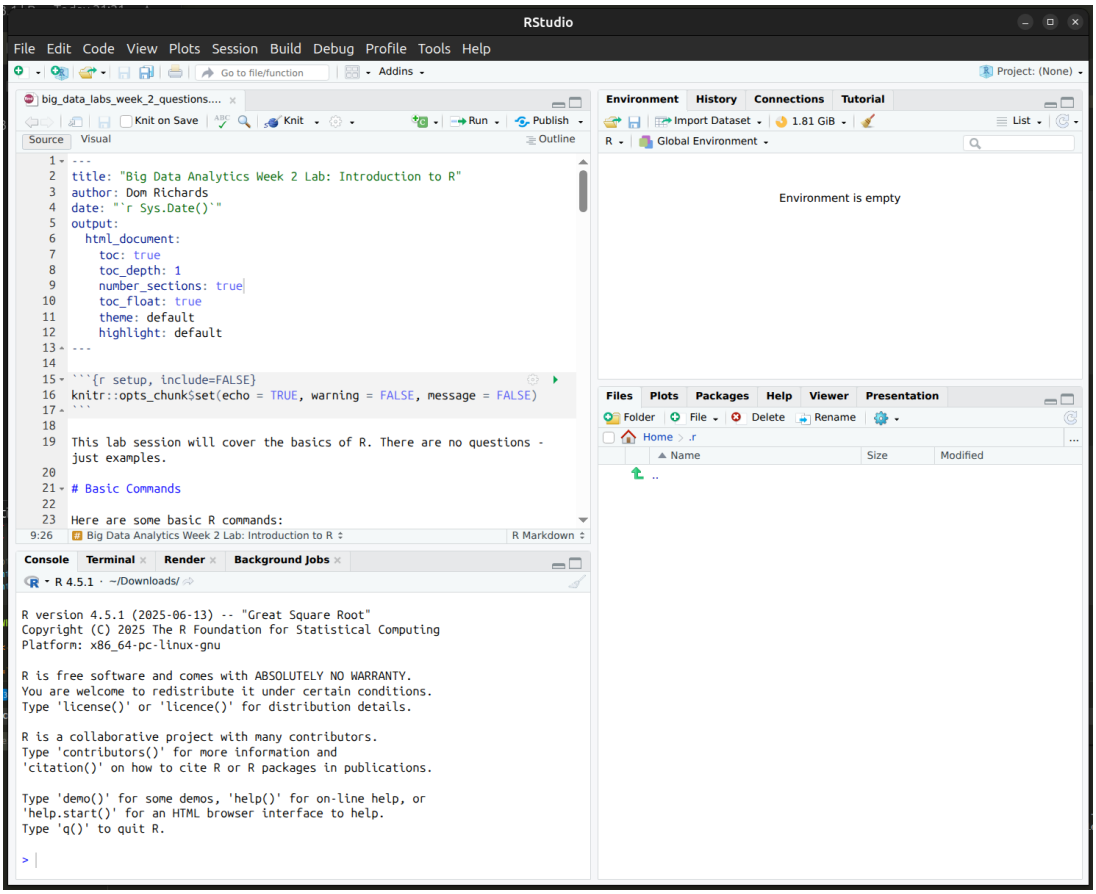


Figure 3: The RStudio Interface showing the four main panes

5 Working with Code Chunks

5.1 Basic Code Chunk Syntax

Code chunks begin with ```{r} and end with ```. For example, the first one is:

````
```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
```
````

The chunk name is `setup` (chunk names have to be unique within the document), and the options are `include=FALSE`. The include option means that neither the code nor its output will be shown in the final document.

5.2 Code Chunk Options

Common chunk options include:

`echo=FALSE` Hide the code, show only results

`eval=FALSE` Show the code but don't run it

`include=FALSE` Run the code but don't show code or results

`warning=FALSE` Suppress warning messages

`message=FALSE` Suppress messages

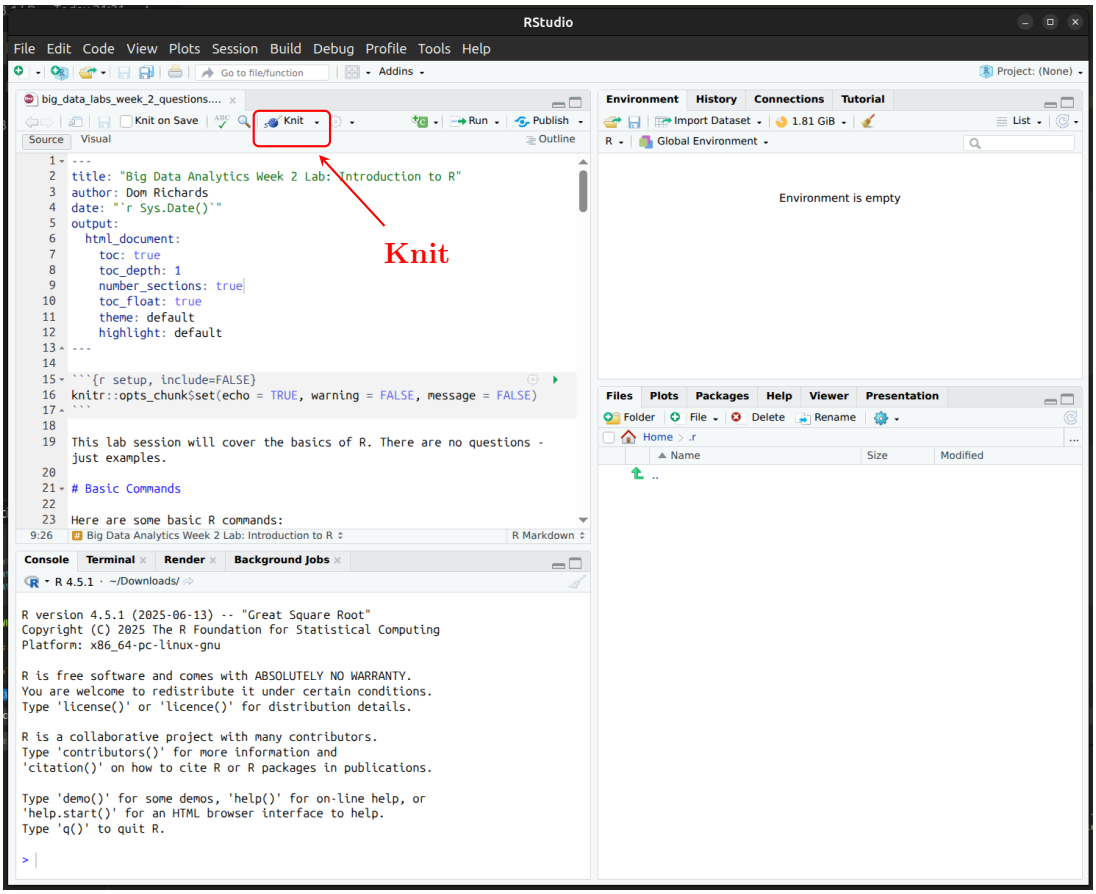


Figure 4: The RStudio Interface showing the knit button location

6 Knitting Your Document

6.1 Knitting to HTML (Default)

The default output format for R Markdown is HTML. To knit your document:

1. Save your R Markdown file (`Ctrl+S` or `Cmd+S`)
2. Click the "Knit" button in the toolbar (shown in figure 4)
3. RStudio will process the document and generate an HTML file
4. The HTML document will open automatically in the Viewer pane or your web browser (figure 5)

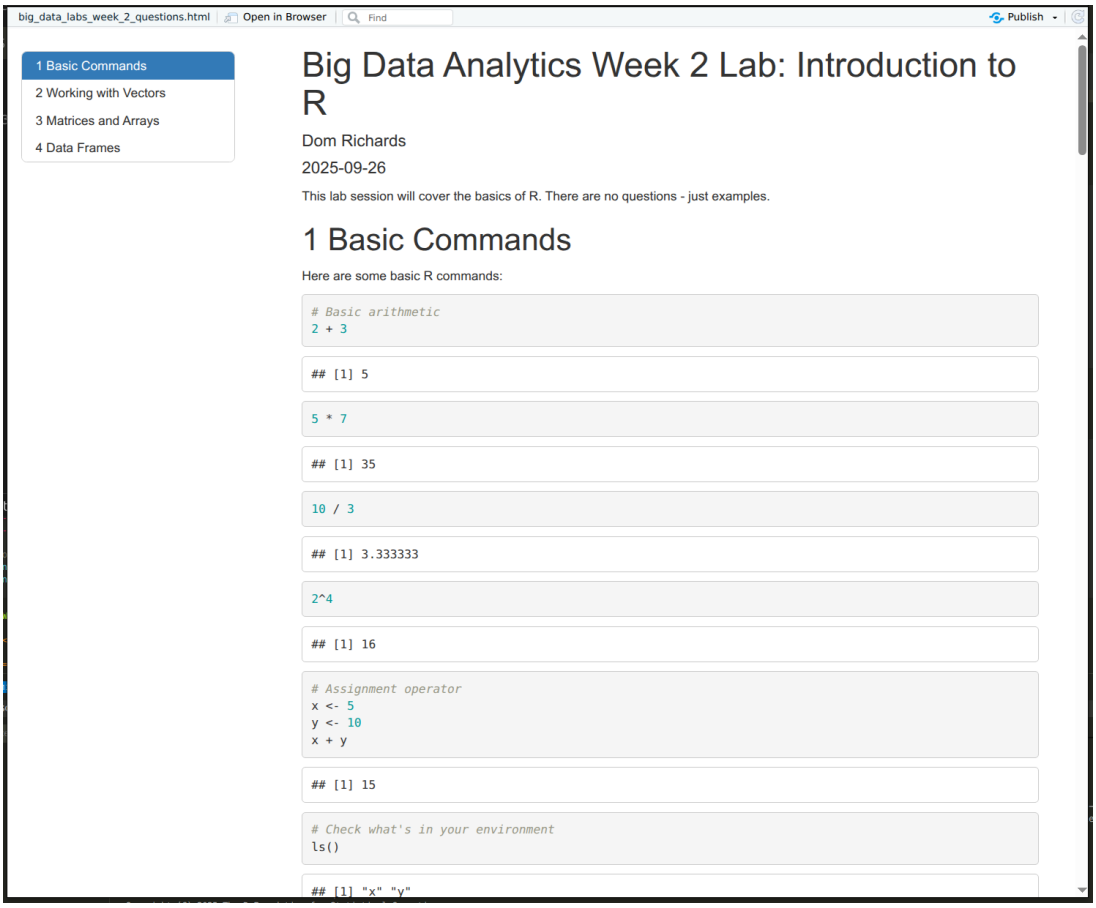


Figure 5: Example of knitted R Markdown document displayed as HTML output